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Studierichting: Informatica
Oefeningenreeks 4A nr. 42.11

$$\int \frac{1-4x}{2x^2+5x+2} dx$$

$$2x^2+5x+2=(2x+1)(x+2)$$

$$\frac{1-4x}{2x^2+5x+2} = \frac{1-4x}{(2x+1)(x+2)} = \frac{A}{2x+1} + \frac{B}{x+2}$$

$$A = \left. \frac{1-4x}{x+2} \right|_{x=-\frac{1}{2}} = \frac{\frac{3}{2}}{\frac{3}{2}} = 2$$

$$B = \left. \frac{1-4x}{2x+1} \right|_{x=-2} = \frac{9}{-3} = -3$$

$$\frac{1-4x}{2x^2+5x+2} = \frac{2}{2x+1} - \frac{3}{x+2}$$

$$\int \frac{1-4x}{2x^2+5x+2} dx = 2 \int \frac{dx}{2x+1} - 3 \int \frac{dx}{x+2}$$

$$= \ln |2x+1| - 3 \ln |x+2| + c$$

References